Additive energy of regular measures in one and higher dimensions, and the fractal uncertainty principle

Terence Tao · 6 December 2020

https://terrytao.wordpress.com/2020/12/06/additive-energy-of-regular-measures-in-one-and-higher-dimensions-and-the-fractional-uncertainty-principle/

tao2typ · converted to Typst 2026-10-14 (UTC)

Laura Cladek and I have just uploaded to the arXiv our paper "Additive energy of regular measures in one and higher dimensions, and the fractal uncertainty principle". This paper concerns a continuous version of the notion of additive energy. Given a finite measure μ on $\mathbf{R}^d$ and a scale $r > 0$, define the energy $\mathrm{E}(\mu, r)$ at scale r to be the quantity $$\mathrm{E}(\mu, r) := \mu^4\big(\{(x_1, x_2, x_3, x_4) \in (\mathbf{R}^d)^4 : |x_1 + x_2 - x_3 - x_4| \leq r\}\big) \quad (1)$$ where μ^4 is the product measure on $(\mathbf{R}^d)^4$ formed from four copies of the measure μ on $\mathbf{R}^d$. We will be interested in Cantor-type measures μ, supported on a compact set $X \subset B(0,1)$ and obeying the Ahlfors-David regularity condition

$$\mu(B(x,r)) \leq C r^\delta$$

for all balls $B(x,r)$ and some constants $C, \delta > 0$, as well as the matching lower bound

$$\mu(B(x,r)) \geq C^{-1} r^\delta$$

when $x \in X$ whenever $0 < r < 1$. One should think of X as a δ-dimensional fractal set, and μ as some vaguely self-similar measure on this set.

Note that once one fixes x_1, x_2, x_3, the variable x_4 in (1) is constrained to a ball of radius r, hence we obtain the trivial upper bound $$\mathrm{E}(\mu, r) \leq C^4 r^\delta. \quad (2)$$ If the set X contains a lot of "additive structure", one can expect this bound to be basically sharp; for instance, if δ is an integer, X is a δ-dimensional unit disk, and μ is Lebesgue measure on this disk, one can verify that $\mathrm{E}(\mu, r) \sim r^\delta$ (where we allow implied constants to depend on d, δ. However we show that if the dimension is non-integer, then one obtains a gain:

> **Theorem 1** If $0 < \delta < d$ is not an integer, and X, μ are as above, then
>
> $$\mathrm{E}(\mu, r) \lessapprox_{C,\delta,d} r^{\delta+\beta}$$
>
> for some $\beta > 0$ depending only on C, δ, d.

Informally, this asserts that Ahlfors-David regular fractal sets of non-integer dimension cannot behave as if they are approximately closed under addition. In fact the gain β we obtain is quasipolynomial in the regularity constant C:

$$\beta = \exp\Big(-O_{\delta,d}(1 + \log^{O_{\delta,d}(1)}(C))\Big).$$

(We also obtain a localised version in which the regularity condition is only required to hold at scales between r and 1.) Such a result was previously obtained (with more explicit values of the $O_{\delta,d}()$ implied constants) in the one-dimensional case $d = 1$ by Dyatlov and Zahl; but in higher dimensions there does not appear to have been any results for this general class of sets X and measures μ. In the paper of Dyatlov and Zahl it is noted that some dependence on C is necessary; in particular, β cannot be much better than $1/\log C$. This reflects the fact that there *are* fractal sets that do behave reasonably well with respect to addition (basically because they are built out of long arithmetic progressions at many scales);

however, such sets are not very Ahlfors-David regular. Among other things, this result readily implies a dimension expansion result

$$\dim(f(X,X)) \geq \delta + \beta$$

for any non-degenerate smooth map $f : \mathbf{R}^d \times \mathbf{R}^d \to \mathbf{R}^d$, including the sum map $f(x,y) := x+y$ and (in one dimension) the product map $f(x,y) := x \cdot y$, where the non-degeneracy condition required is that the gradients $D_x f(x,y), D_y f(x,y) : \mathbf{R}^d \to \mathbf{R}^d$ are invertible for every x, y. We refer to the paper for the formal statement.

Our higher-dimensional argument shares many features in common with that of Dyatlov and Zahl, notably a reliance on the modern tools of additive combinatorics (and specifically the Bogulybov-Ruzsa lemma of Sanders). However, in one dimension we were also able to find a completely elementary argument, avoiding any particularly advanced additive combinatorics and instead primarily exploiting the order-theoretic properties of the real line, that gave a superior value of β, namely

$$\beta := c \min(\delta, 1-\delta) C^{-25}.$$

One of the main reasons for obtaining such improved energy bounds is that they imply a *fractal uncertainty principle* in some regimes. We focus attention on the model case of obtaining such an uncertainty principle for the semiclassical Fourier transform

$$\mathcal{F}_h f(\xi) := (2\pi h)^{-d/2} \int_{\mathbf{R}^d} e^{-ix\cdot\xi/h} f(x)\, dx$$

where $h > 0$ is a small parameter. If X, μ, δ are as above, and X_h denotes the h-neighbourhood of X, then from the Hausdorff-Young inequality one obtains the trivial bound

$$\left\| 1_{X_h} \mathcal{F}_h 1_{X_h} \right\|_{L^2(\mathbf{R}^d) \to L^2(\mathbf{R}^d)} \underset{C,d}{\lesssim} h^{\max(\frac{d}{2}-\delta, 0)}.$$

(There are also variants involving pairs of sets X_h, Y_h, but for simplicity we focus on the uncertainty principle for a single set X_h.) The *fractal uncertainty principle*, when it applies, asserts that one can improve this to

$$\left\| 1_{X_h} \mathcal{F}_h 1_{X_h} \right\|_{L^2(\mathbf{R}^d) \to L^2(\mathbf{R}^d)} \underset{C,d}{\lesssim} h^{\max(\frac{d}{2}-\delta, 0)+\beta}$$

for some $\beta > 0$; informally, this asserts that a function and its Fourier transform cannot simultaneously be concentrated in the set X_h when $\delta \leq \frac{d}{2}$, and that a function cannot be concentrated on X_h and have its Fourier transform be of maximum size on X_h when $\delta \geq \frac{d}{2}$. A modification of the disk example mentioned previously shows that such a fractal uncertainty principle cannot hold if δ is an integer. However, in one dimension, the fractal uncertainty principle is known to hold for all $0 < \delta < 1$. The above-mentioned results of Dyatlov and Zahl were able to establish this for δ close to $1/2$, and the remaining cases $1/2 < \delta < 1$ and $0 < \delta < 1/2$ were later established by Bourgain-Dyatlov and Dyatlov-Jin respectively. Such uncertainty principles have applications to hyperbolic dynamics, in particular in establishing spectral gaps for certain Selberg zeta functions.

It remains a largely open problem to establish a fractal uncertainty principle in higher dimensions. Our results allow one to establish such a principle when the dimension δ is close to $d/2$, and d is assumed to be odd (to make $d/2$ a non-integer). There is also work of Han and Schlag that obtains such a principle when one of the copies of X_h is assumed to have a product structure. We hope to obtain further higher-dimensional fractal uncertainty principles in subsequent work.

We now sketch how our main theorem is proved. In both one dimension and higher dimensions, the main point is to get a preliminary improvement $\mathrm{E}(\mu, r_0) \leq \varepsilon r_0^\delta$ (3) over the trivial bound (2) for any small $\varepsilon > 0$, provided r_0 is sufficiently small depending on ε, δ, d; one can then iterate this bound by a fairly standard "induction on scales" argument (which roughly speaking can be used to show that energies $\mathrm{E}(\mu, r)$ behave somewhat multiplicatively in the scale parameter r) to propagate the bound to

a power gain at smaller scales. We found that a particularly clean way to run the induction on scales was via use of the Gowers uniformity norm U^2, and particularly via a clean Fubini-type inequality

$$\|f\|_{U^2(V\times V')} \leq \|f\|_{U^2(V;U^2(V'))}$$

(ultimately proven using the Gowers-Cauchy-Schwarz inequality) that allows one to "decouple" coarse and fine scale aspects of the Gowers norms (and hence of additive energies).

It remains to obtain the preliminary improvement. In one dimension this is done by identifying some "left edges" of the set X that supports μ: intervals $[x, x+K^{-n}]$ that intersect X, but such that a large interval $[x-K^{-n+1}, x]$ just to the left of this interval is disjoint from X. Here K is a large constant and n is a scale parameter. It is not difficult to show (using in particular the Archimedean nature of the real line) that if one has the Ahlfors-David regularity condition for some $0 < \delta < 1$ then left edges exist in abundance at every scale; for instance most points of X would be expected to lie in quite a few of these left edges (much as most elements of, say, the ternary Cantor set $\{\sum_{n=1}^\infty \varepsilon_n 3^{-n} \varepsilon_n \in \{0,1\}\}$ would be expected to contain a lot of 0s in their base 3 expansion). In particular, most pairs $(x_1, x_2) \in X \times X$ would be expected to lie in a pair $[x, x+K^{-n}] \times [y, y+K^{-n}]$ of left edges of equal length. The key point is then that if $(x_1, x_2) \in X \times X$ lies in such a pair with $K^{-n} \geq r$, then there are relatively few pairs $(x_3, x_4) \in X \times X$ at distance $O(K^{-n+1})$ from (x_1, x_2) for which one has the relation $x_1 + x_2 = x_3 + x_4 + O(r)$, because x_3, x_4 will both tend to be to the right of x_1, x_2 respectively. This causes a decrement in the energy at scale K^{-n+1}, and by carefully combining all these energy decrements one can eventually cobble together the energy bound (3).

We were not able to make this argument work in higher dimension (though perhaps the cases $0 < \delta < 1$ and $d-1 < \delta < d$ might not be completely out of reach from these methods). Instead we return to additive combinatorics methods. If the claim (3) failed, then by applying the Balog-Szemeredi-Gowers theorem we can show that the set X has high correlation with an approximate group H, and hence (by the aforementioned Bogulybov-Ruzsa type theorem of Sanders, which is the main source of the quasipolynomial bounds in our final exponent) X will exhibit an approximate "symmetry" along some non-trivial arithmetic progression of some spacing length r and some diameter $R \gg r$. The r-neighbourhood X_r of X will then resemble the union of parallel "cylinders" of dimensions $r \times R$. If we focus on a typical R-ball of X_r, the set now resembles a Cartesian product of an interval of length R with a subset of a $d-1$-dimensional hyperplane, which behaves approximately like an Ahlfors-David regular set of dimension $\delta - 1$ (this already lets us conclude a contradiction if $\delta < 1$). Note that if the original dimension δ was non-integer then this new dimension $\delta - 1$ will also be non-integer. It is then possible to contradict the failure of (3) by appealing to a suitable induction hypothesis at one lower dimension.

Comments (10)

rafik zeraoulia · 6 December 2020

Dear Prof Tao ,In which case we could have a uniforme distribution (Gaussian) for additive energy ? for integer dimension or non integer ?

Anonymous · 7 December 2020

Proposition 2.1: The last expression is missing; "In fact we can take ..."
[This will be deleted in the next revision of the ms – T.]

Anonymous · 7 December 2020

What was the motivation behind the introduction of the (apparently artificial) expression $x_1 + x_2 - x_3 - x_4$ in the definition (1) ?

> **Terence Tao** · 7 December 2020
>
> Additive energy is a statistical measure of the extent to which a given set X (in this case, the support of μ) contains additive structure, which roughly speaking means that the set X behaves approximately as if it is closed under addition. The intuition here is that if X is approximately closed under addition, then the sums $x_1 + x_2, x_1, x_2 \in X$ range in an unusually small set, so one should expect to see a lot of "collisions" $x_1 + x_2 = x_3 + x_4$ or "near-collisions" $x_1 + x_2 = x_3 + x_4 + O(r)$" between these sums. The additiv energy measures these collisions, and turns out to be a convenient proxy for additive structure for a number of reasons, such as its connection to the Gowers uniformity norm and to the Fourier transform of the underlying measure μ.

Rafik Zeraoulia · 7 December 2020

if am true I think this definition for additive energy coincide with sub-energy function definition such that one can construct sub energy function in one dimension or higher by the following assumption : f_c(x_1,x_2,x_3,x_4) go to infinity if x_1+x_2+x_3+x_4 go also to infinity one can replace this weaker condition x_1+x_2+x_3+x_4 with stronger conditions like x_1 go infty or x_2 go infty ,.... ,Now my question here is : Is that Additive energy function a physical notion or mathematical notion like sub-energy function which it used widely in statistical mechanics ?

Xiao · 7 December 2020

A typo just founded p. 34 "Let u' denotes"

Anonymous · 7 December 2020

In Proposition 5.1 a symbol may be missing in the sentence 'In fact one can take to be quasipolynomial in (...)' after the word take. Is it r0 ?
[Thanks, this will be corrected in the next revision of the ms. -T]

Anonymous · 8 December 2020

In the bibliography, the reference #13 could be quoted as follows:
[13] O.E. Raz, J. Zahl. . 29 pages, arXiv:2010.04845, October 2020.

Anonymous · 8 December 2020

Instead of 'Theorems 1.4, 1.5' you could mention 'Theorems 1.4 and 1.5' (3 occurrences)

Anonym · 8 December 2020

Dear Professor Tao,

Very interesting and unexpected paper.

In order to optimize it, I just have a very minor lexical suggestion related to the expressions compare and compared “against” or “to”.

Maybe these expressions should be replaced by compare and compared “with”.

Yours sincerely